

EE 435

KEZ

CHAPTER ONE:

A Glimpse at RF and Wireless Technology

1.0 Introduction

The very last 20 years were the most exciting in telecommunication evolution. Cellular phone communications, in particular, have evolved from a small market to an integral part of everyday life. Today's wide area and local area wireless systems are positioned to take us to the other level of voice and data services, in which all of the Internet's functionality is basically at our disposal anywhere we go, thanks to reasonable progressions in computing and information technology.

We examine the history of wireless technology in this chapter, study the current network environment, and incorporate a few of the cutting-edge aspects discussed subsequently in this book.

1.1 Wireless Communication: A Brief History

Figure 1.1 illustrates the progression of telecommunications over time. The radio transmission industry is still in its 2nd century. Faraday, Maxwell, Hertz, and many others' ground-breaking work in the nineteenth century gave rise to Marconi's broadband telegraph somewhere at beginning of the twentieth century. Ever since the portable voice radio systems throughout the early 20th century, there were predecessors to radio communication as we recognize it. Broadcasting technology advanced in the following decades, with the introduction of telecast tv and radio, as well as handheld multiband walkie-talkies during the Second World War. Cellular innovation was introduced in the 1940s, also with the potential to split wireless radio networks into "cells" to prevent collisions and increase capacity. That's the foundation for today's modern mobile broadband wireless LANs and vast area of sound innovations. Satellites were shot into space as connectivity relays after a few years after the first

communications satellite in 1957.

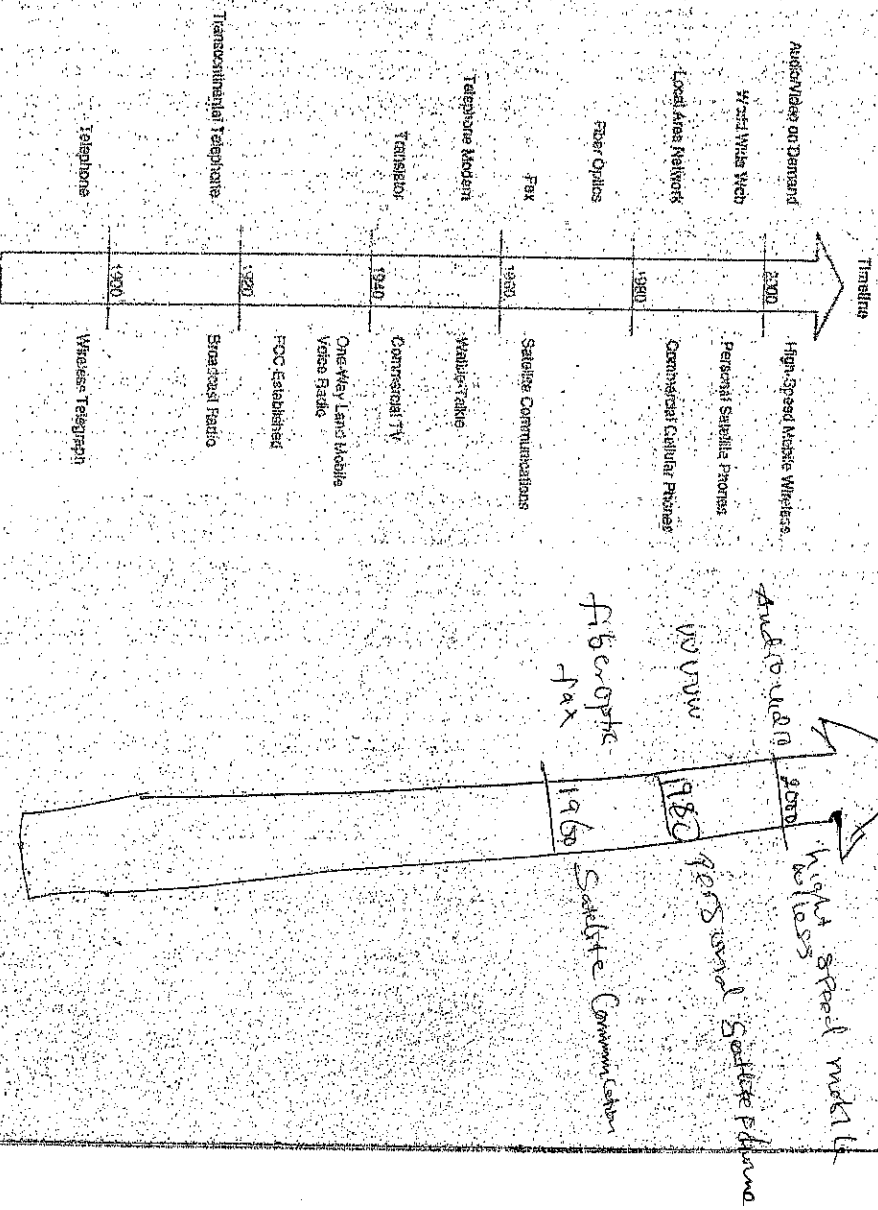


Figure 1.1 The chart displays general advancements in mobile communications on the left as well as wireless-specific advancements on the right.

The Federal Communications Commission (FCC) allocate segments of the frequency band to cellular telephones in 1969. The numeric paging code was standardized by the Post Office Code Standardization Advisory Group (POCSAG) in the late 1960s, and AT&T introduced the very first cellular phone infrastructure using a cellular system.

The Federal Communications Commission (FCC) approved the use of emerging innovations in the 800 MHz mobile broadband spectral region in 1987, and then the Second-generation wireless communications (code division multiple access [CDMA], time division multiple access [TDMA], and global system for mobile communication [GSM]) were assessed united states pretty soon after. Additional

Innovations such as voicemail messages, telex, and instant messengers have been supported as a result of the adoption of digital technologies.

The increase in wireless utilization in the late 1990s (coinciding with the Internet boom) resulted in nearly broadband wireless voice calls across Nigeria and much of the rest of the region. Private wireless data capabilities, as shown by protocols including SMS, WAP, ReFlex, Bluetooth, i-Mode, and 802.11, provide a spectrum of mobile data which are not much behind.

1.2 Wireless Systems in Use

Wireless transmission has indeed been widely employed ever since the days of wireless telegraphy. Ever since modern civilization has indeed been striving for more telecommunication portability. Originally, wireless communication was confined to a single pair of users of a particular channel pair. The movement distance was determined by the transmitter power, aerial type, and operating frequency. As the number of users increased, it became increasingly difficult to accommodate them within the restricted accessible frequency band. To address this issue, the notion of cellular communication was developed. Cellular communication nowadays is based on a fundamental unit known as a cell. A cell is made up of a tiny hexagonal region with a base station in the middle that connects with the subscriber.

To allow for numerous users it is possible to employ Time Division Multiple Access (TDMA), Code Division Multiple Access (CDMA), Frequency Division Multiple Access (FDMA), and respective variants. A variety of mobile wireless systems, such as AMPS, PACS, GSM, NTT, PHS, and IS-95, have now been employed in different locations, each employing a unique set of bands and assigning a unique percentage of subscribers and channels.

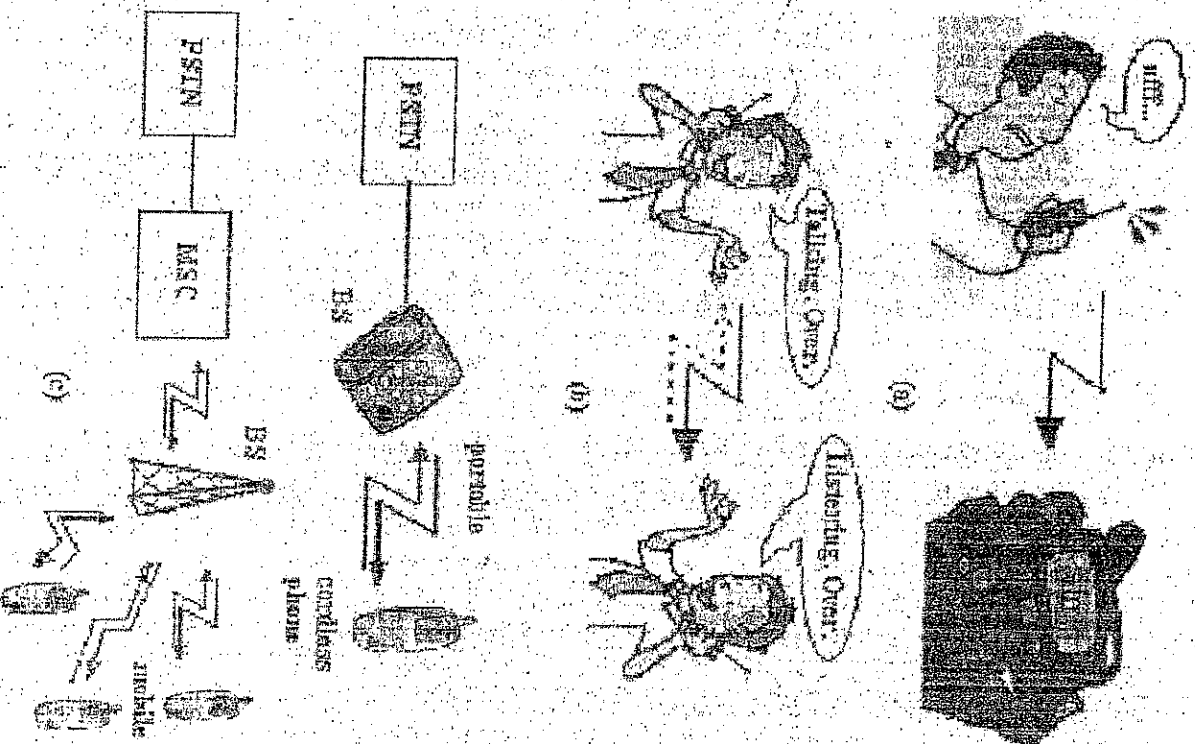


Figure 1.2 The three fundamental radio transmission strategies: (a) simplex, (b) half-duplex, and (c) full-duplex.

1.3 Basic Concept of Wireless networks

Before actually delving into unique wireless technologies, it's a good idea to go through the key features of a general radio communication system.

Figure 1.3 depicts a wireless system with a signal being transmitted from the transmitter on the left to the receiver in the middle. Other features of the surroundings are depicted to demonstrate the difficulties associated with wireless communications. Many studies are being conducted to address these issues in order to improve Radiofrequency connectivity.

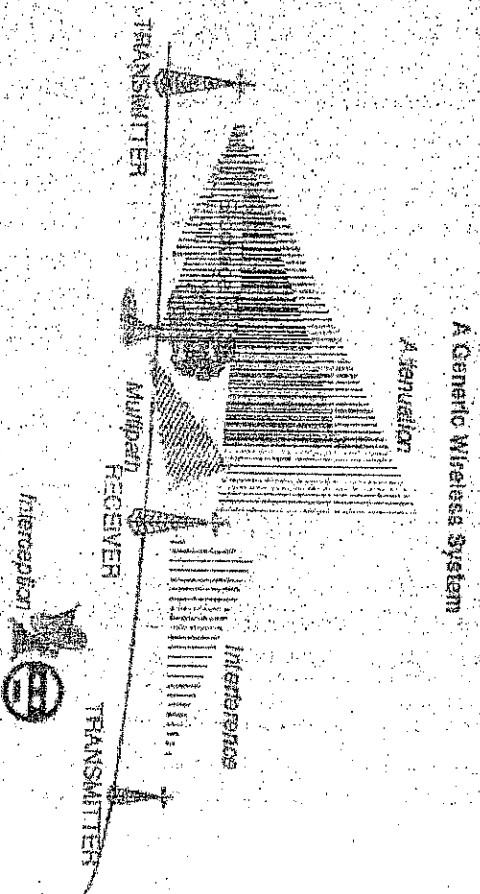


Figure 1.3 *A standard wireless system. The cellular format's intrinsic shortcomings are countered by its adaptability and movement assistance.*

Furthermore, radio transmissions are susceptible to reflection, thus resulting in multipath fading, another type of transmission loss. Because a reflected wave propagates at a greater distance between transmitter and receiver, it reaches the receiver with such a time lag compared to a transmission that follows a straight (or lesser reflected) route. Two associated signals that are timed apart may neutralize each other.

An additional challenge for the receiver working in an unsecured wireless surrounding is the potential of many other comparable signals utilizing the same channel and getting to the receiver at the

same time as the intended signal, thus producing disruption. Lastly, the unsecured medium allows for eavesdropping or interruption, which occurs when a third party catches the sent signal without authorization, thereby jeopardizing the confidentiality of the system's subscribers.

Each of the issues depicted in Figure 1.3 describes a situation in which wireless technology is inferior to a similar landline communication technology. So, why are wireless technologies so common? Each wireless deployment may have its unique design justification, but two basic explanations cover the vast majority of scenarios. For starters, wireless technologies allow for greater deployment options. Because no physical linking media is required, a radio link may be orders of magnitude quicker than a wireline connection, regardless of whether connecting a laptop computer in a conference room or a pipeline monitor in the Arctic. Secondly, wireless technologies can provide you the freedom to move anywhere. Many applications benefit from, if not need, mobile communication, which is simply not effectively supported by a wired connection.

It's worth noting that most of these "deficiencies" may be intelligently used to the subscriber's benefit. Absorption and obstructing, for instance, can be used to reduce signal transmission and eavesdropping chances.

After delving too deeply into the benefits and drawbacks of wireless, now let's revert to the topic of how the wireless landscape is inhabited, beginning with fixed wireless systems and progressing to the local area and wide area networks.

1.3.1.1 Systems with Fixed Locations

Fixed wireless technologies, by definition, can handle high-gain immovable antennas and power grid interconnections; the consequent high signal-to-noise ratios allow for high bandwidth over long-distance line-of-sight pathways. The next sections explain several types of fixed systems, including Radiofrequency and optical systems, as well as terrestrial and space-based mechanisms.

1.3.1.1.1 Point-to-Point Radio Frequency

High-speed megabit (T-1, T-3) telecom lines are often replaced by point-to-point microwave networks. They usually use subscriber-end

equipment and licensed frequencies over 2 GHz. They employ highly directional antennas with a range of up to 10 miles if the line of sight is available. (Systems are commonly installed atop big buildings and mountaintops to enhance the line of sight.) Unlicensed equipment running in the 2.4 GHz industrial, scientific, and medical (ISM) or 5 GHz unlicensed national information infrastructure (UNII) bands have become increasingly popular in this area.

→1.3.1.2 Point to Multipoint

The multichannel multipoint distribution service (MMDS) and the local multipoint distribution system (LMDS) are carrier-grade solutions for wireless Connectivity and basic telecommunications services. These systems have been given spectrum in various super high frequency (SHF) bands, with a throughput of tens of megabits per second. The 802.16 working group of the Institute of Electrical and Electronic Engineers (IEEE) is creating a family of standards for wireless metropolitan area networks in this vein (MANs). The technology competes with cable modem and DSL connectivity, but with more mobility. The present telecommunications crisis has hampered the rollout of these technologies.

→1.3.1.3 VSAT as Well as other Geostationary Communications Satellites

The very small aperture terminal (VSAT) principle is similar to stationary terrestrial multiple access networks, just that a satellite transmitter is utilized rather than a tower- or rooftop-mounted base station. At 22,236 miles altitude, a limited fraction of geosynchronous earth orbit (GEO) satellites revolves around the equator, keeping a stable location in reference to a location on the globe. These transceivers could be used for bidirectional transmission relay with wide throughputs, such as transmitting data, Internet connections, and audio/video signal dissemination. The base station antenna is permanent throughout the operation, but it is quickly deployable, making it perfect for backup and recovery as well as other short-term circumstances.

Multiple licensed bands are used once again. Full satellite channels are frequently segmented using different access mechanisms and resold at speeds ranging from 20 kbps to 2 Mbps. According to the

Global VSAT Forum³, approximately 500,000 VSAT stations are in use across the globe.

→1.3.1.4 Optical in Free Space

Free space optical (FSO) technologies are a relatively new addition to the industry, sharing features with both point-to-point Radiofrequency and optical fiber technology. A concentrated signal, similar to point-to-point systems, transports greater bitstreams among two sites; similarly, to fiber optics, a beam of light-emitting diode (LED) is used to create the optical signal that transports the information.

FSO, which operates without a spectrum license and has rates in excess of 1 Gbps, is an appealing option for several backhauls and LAN extension purposes. The sensitivity of today's systems to optical blocking, especially from fog, is indeed one of the flaws.

→1.3.2 Local Area Networks

Local area systems, as opposed to fixed systems, achieve the highest throughput by closeness. They often enable some range of motion on the part of the user, allowing for more adaptable operation circumstances. The technology solutions under consideration do not necessitate a frequency band license.

→1.3.2.1 Infrared

Infrared transmissions are used in a variety of simple algorithms and protocols, including television remote controls. In order to promote a huge spectrum of mode selection, the Infrared Data Association (IrDA) has formalized multimegabit-per-second transmission of data. These are usually exceedingly, relatively brief (i.e., inches to feet of detachment among units) apps available that require a significant amount of positional accuracy among transmitter and receiver. According to IrDA4, this innovation is supported by over 300 million pieces of equipment.

→1.3.2.2 Personal Area Networks (PANs) on Bluetooth

Bluetooth (named after Harald Bluetooth, an ancient Danish Viking king) operates at a 1 Mbps broadcast rate in the unauthorized 2.4 GHz ISM band and is designed for minimum cost and maximum interconnectivity. It's known as a personal area network (PAN) or

wire substitute technique, and it's a standard in mobile phones, computers, PDAs, and printers, among other product lines that currently rely on cables for connectivity. In fields as diverse as teleconferencing and automotive assistance, so many software packages (as described by Bluetooth "Profile" standards) will become available over time.

1.3.2.3 Wireless Local Area Networks (WLANs)

Wireless local area networking (WLAN) innovations have been the big news in wireless technology in the last year or so, instead of the much 3rd generation (3G) cellular. Such techniques (mainly centered on the IEEE 802.11 family of standards and commercialized as Wi-Fi) function in the unlicensed band and extend or substitute cabled LANs. The Wi-Fi technology, popularly known as Wireless Ethernet, enables conventional programs to run invisibly across a transparent layers 1 and 2.

Originally used as workplace LAN expansions, their low cost, fast speed (to tens of Mbps), and simplicity of use have rendered them attractive in the residential office setting and, to a smaller degree, as a replacement for classic point-to-point high-frequency applications. Additional vertical uses are beginning to emerge. Several toll collecting systems in vehicles use 802.11, and commercial flights have declared the deployment of WLANs in their planes for in-flight passenger usage. Unlimited access to internet connections is cropping up in a variety of high-tech locations, posing a threat to the business models of existing cellular providers.

1.3.3 Wide Area Networks

In practice, wide-area mobile systems forfeit throughput in exchange for extended propagation pathways, mobility management, reduced device footprint, and higher coverage. Wide area systems include paging and cellular network connectivity.

1.3.3.1 Paging

For many years, paging provided a dependable delivery system for short text sequences including telephone numbers. Outstanding dependability was accomplished by simplex data transfer, strategically located high-power transmitters, and strong encoding. To contend with emerging cell phone services in the nineties, several

paging networks were modified to enable two-way transmissions and bigger message payloads, rendering them functionally equivalent to the terrestrial packet systems discussed later in this chapter.

1.3.3.2 Satellite Services for Mobile Devices

Several low and medium earth orbit satellite systems were designed in the nineties. Just a handful of these have been ever deployed, and those that were reportedly struggled to sustain economic viability in the midst of ever-increasing domestic cell phone coverage.

Productive telecommunications companies gain from mobile satellite communications in niche markets. Inmarsat connects marine and other high-value assets via a GEO satellite network. Omni-Tracs supplies long-haul transportation companies with vehicle monitoring and satellite systems.

1.3.3.3 Private Specialized Systems

The majority of wide-area radio transmissions, such as cellular and paging, are supplied as public infrastructure by carriers. Military systems are the exception, as they often utilize customized patent architectures in specified frequency ranges. Specialized private systems, such as those used for audio dispatch and remote asset monitoring, are also another instance. These schemes use authorized radio frequencies and are made up of multiple mast sites as well as a large number of fixed or mobile subscriber devices that are managed by a company or government. SMR (specialist mobile radio), MAS (multi-address system), trunked radio, PMR (private mobile radio), and other terms are used to describe them. Dedicated radio systems, lacking the economies of scale of their mass-market counterparts, are frequently a decade behind the similar device, having most still using analog modulation, for instance, long after cellular has transitioned to digital.

1.3.3.4 Cellular

With a constrained frequency band, i.e., a finite number of channels, mobile telephony networks must support a large customer base over a vast geographic region. If only a singular transmitter/receiver is utilized with a base station, there would not be enough power available at a considerable distance from the Base station. As a result,

a high-powered transmitter is required for wide geographic network coverage.

A high-powered radio transmitter, on the other hand, causes environmental damage. As a result, mobile communication necessitates the use of low-power transmitters and the division of the service area into small parts known as cells. Each cell utilizes a subset of the available channels, whereas a cluster of nearby cells utilizes all of them. The *cluster* is a type of group like this.

Because this cluster may replicate itself, the same collection of channels can be utilized repeatedly. Every cell is equipped with a low-power transmitter with a network coverage corresponding to the cell's size. The foundation of the cellular idea is the approach of replacing a particular high transmitter with numerous low-powered transmitters to accommodate a lot of users.

1.3.4 Applications

We recently spoke about the present wireless situation from a technology standpoint. We may also think of wireless goods in terms of the capabilities they support such as telephony, messaging, private data, and Internet service.

1.3.4.1 Voice

The far more visible wireless voice technology is cellular. Lateral voice markets are supported by SMR and communications satellites, and different technologies (like Bluetooth) are employed for short-range wireless communication. Attempts are being made to enable voice calls using WLAN technology.

1.3.4.2 Messaging

Small wireless communications have mirrored the prevalence of e-mail in a cabled context. The majority of traffic on the Mobitex, Ardis (packet services), and ReFlex (paging) systems today is message data. Texting, which transports text messages of up to 150 characters across the mobile network, is supported by terrestrial and satellite voice networks. On certain networks, its replacements, enhanced messaging service (EMS) and multimedia messaging service (MMS service), now include upgraded material such as photos.

3.4.3 Access to the Internet

Prior earlier, wireless Internet service entailed a 10 kbps wireless channel and a restricted web view through the SMS wireless application protocol (WAP) or a related technique. Current online mobile and wireless Ethernet solutions provide customers with complete access to the web at requiring relatively across a vast geographic region. VSAT capabilities and, in some regions, fixed multiple access systems can be found for a greater rate of stable access.

1.4 Future trends in wireless technology.

The rapid growth of the Internet, including its digital products and services, has posed additional challenges for such advancement of mobile communications networks. Significant advancements in modulation techniques will soon raise wireless Internet connection speeds from the present 1.8 Mbps to more than 10 Mbps. Bluetooth is quickly becoming a standard function in cell devices for making local connections.

Due to technological advancements and increasing competition between many telecom operators, mobile communication had also provided people with global connectivity at a cheaper price. Furthermore, several active research fields have the potential to bring quantum advancements to the wireless environment.

1.4.1 Software or Cognitive Radio

Open system interconnects (OSI) Layer 1 (physical) is implemented in (mainly analog) hardware in conventional radios, including filters and amplifiers, for instance, built for functioning at specified frequencies compatible with the radio's original purpose. Layer 2 (data connection) is often accomplished in specialized digital hardware or an overall central processing unit; subsequent layers are typically operated by software/firmware in general-purpose microchips.

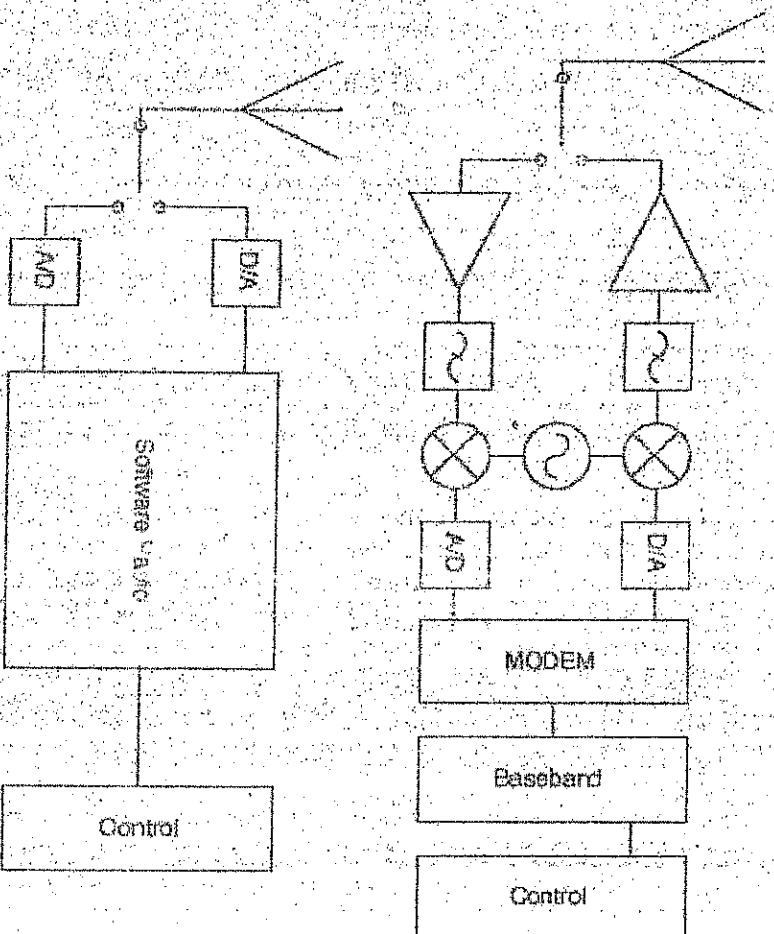


Figure 1.4: Software Radios

The top half of Figure 1.4 depicts the block diagram of traditional radio.

The objective behind software radio is really to migrate much many radio functions (including the physical layer) into software as reasonable. However, achieving the increased sampling and signal processing necessary for a software-based radio is still to be shown economically viable. Lately, a similar example of software-defined radio has surfaced as a research and development subject. Under this scenario, dynamically reconfigurable logic arrays perform computationally complex features like source code or encrypted communications as required. Upon receiving, a logic block could be set like a decoder, and subsequently changed like an encoder during the following send phase. Figure 1.4's lower section depicts a block schematic of a software-defined radio.

The benefits of software radio and software-defined radio are similar:

- Convenience: The radio does not have to be developed again with exact features of the planned usage in mind.
- Completely renewable: A fresh software installation is all that is required to begin using new radio technology.
- Price: Digital logic gates are intrinsically less costly than passive components while using modern technology.
- Usage of energy: For a software/digital operation, there could be greater opportunities to improve for reduced power consumption (thus better battery life) than with an analog feature.

1.4.2 Ultrawideband Signals

Ultra wide band (UWB) has now been suggested as a technique for a broad range of applications, including high-speed connectivity, position detection, and brief earth imaging. UWB, with pulse modulation and bandwidths in addition to 1 GHz, has the ability to expand the benefits of the spread spectrum still further. The intention is for UWB transmitters to function in an unlicensed state that overlaps legal frequencies. The FCC established guidelines enabling minimal UWB operation following extensive research, but uncertainties persist in the thoughts of spectrum licensees worried regarding interference.

1.4.3 Antenna arrays with Intelligence

For decades, "smart" antennas have indeed been utilized in specialized applications, particularly by the army. Plans to develop economically appealing technologies for uses including cellular base stations and wireless LAN hot spots are ongoing. Smart antennas, also known as phased array antennas, are an advanced kind of antenna variety. An antenna array is made up of a group of separate antenna components, each one may receive (as well as transmit) signals. When a signal arrives at the array, it is normally collected at every component with a tiny temporal delay. The antenna design is efficiently concentrated on the remote transmitter by dynamically modifying the phase (time delay) for every received signal and afterward aggregating. Signals entering from these other angles are not amplified by the aggregating and are lost in the noise. The same approach can be used to improve both the broadcast and received

signals. One of the technological hurdles is to constantly alter the deviations in order to retain concentration on a distant mobile terminal. A simplified example of the smart antenna operating is depicted in Figure 1.5.

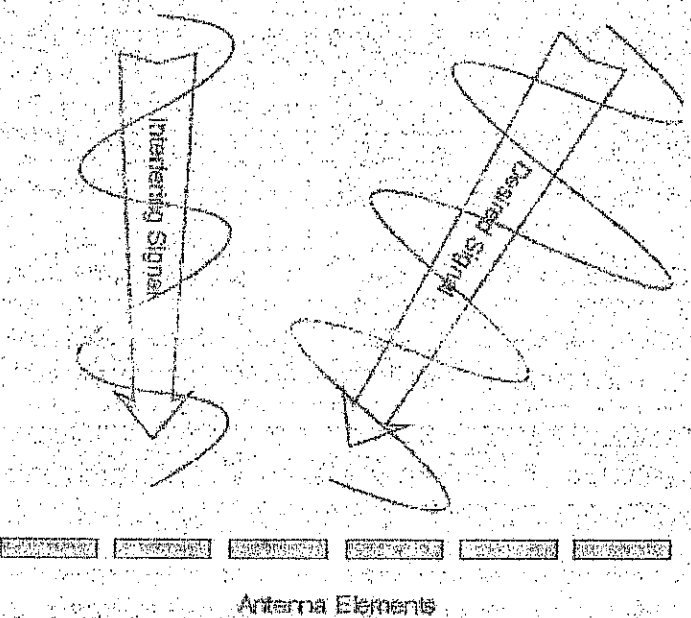


Figure 1.5 Smart antenna.

Figure 1.5 depicts the collection of intended and interfering signals at every antenna array. The displacement variants of the resonator external forces cancel each other whenever the received signals are aligned to optimize the intended signal.

1.4.4 Sophisticated Connectivity

A percentage of advancements at OSI layers 2 and well above would then profit wireless subscribers, with just some instances being discussed below. In the last two decades, the Internet protocol (IP) has developed to the point where connecting from any pc to almost every other pc all around the globe has become a priority, and it can be done efficiently and rapidly. This is also valid regarding phone line communication. In certain ways, mobile communications are more sophisticated than IP networking since it enables individuals to accept a call to his or her distinct identity (telephone number, such as the

home area code) while traveling to a different market with such a new area code. This is no longer applicable with data networking, where a computer's distinctive identification (IP address) is associated with a specific "position" on the network (i.e., a specific router). As the client reconnects via a separate router, the problem is resolved by allocating a new IP address. Some apps are hampered by this, as they are unable to stay connected when their enabling hardware travels.

A collection of characteristics referred to as mobile IP provides a solution to these issues by offering services to cellular data consumption that are equivalent to those offered by the mobile telephone network.

As previously stated, one possible shortcoming of wireless communication systems is cybersecurity. First-generation mobile networks were susceptible to service fraud and also eavesdropping. Such omissions were remedied by including encryption and authentication technologies in second-generation technologies. Similarly, standards will soon fix the security vulnerabilities of early IEEE 802.11 wireless Local - area networks.

Another important finding of this study would be the quality of service quality (QoS). When confronted with the fluctuating latency, channel defects, and throughput intrinsic in several wireless connections, programs (e.g., audio or video) which perform effectively on dedicated cable links sometimes have difficulty. Broadband Quality of service capabilities will give a much more consistent environment for such programs to run upon.

1.4.5 Enhanced Throughput

Except for a few limited particular applications like telemetry, each wireless communication is always striving for increasing broadband speeds. Additional data may be squeezed into usual channels using advanced modulation and encoding. Furthermore, there is indeed a steady shift to high frequency, which have more capacity and hence the possibility for increased throughput.

For instance, contrast 802.11a, which provides 54 Mbps on every one of up to 12 concurrent channels at 5 GHz, to 802.11b, which enables 11 Mbps on every one of three main streams at 2.4 GHz. According to the latest press reports, the extra band in the 5 GHz band is being

suggested to enhance what has already been approved for unauthorized usage.

1.4.6 Improved Access

We've come to anticipate near-universal connectivity from the cell phone providers. This was accomplished via network tuning and significant investment in communications infrastructure. Succeeding technologies and company restructurings will enable even broad global roaming. Whereas there are now multiple quasi-mobile technologies, there are just two competitors for significant worldwide rollout in the fourth generation, and mobile providers are insuring a far higher level of interoperability among technologies.

Furthermore, the growing demand for public and private Wi-Fi hot spots provides cellular data consumers with some other connection choices. Bluetooth clients transport their network connection (PAN) everywhere they go. Microwave technologies are also projected to provide connection possibilities for those hard-to-reach distant locations.

1.4.7 Widespread availability

Not only is network connectivity expanding, but so, too, is the proliferation of wireless devices. Because radios have shrunk in size and cost, they can now be added to practically any device. Improvements in the current digital era, and also wireless devices, have made radios practical in places in which they were previously not.

We are already seeing broadcast PDAs, cameras, smartwatches, and automobiles in addition to cordless devices and personal computers. Wireless communication is possible with any equipment that has an integrated microchip. Figure 1.6 depicts the increase of mobile network access devices in Nigeria.

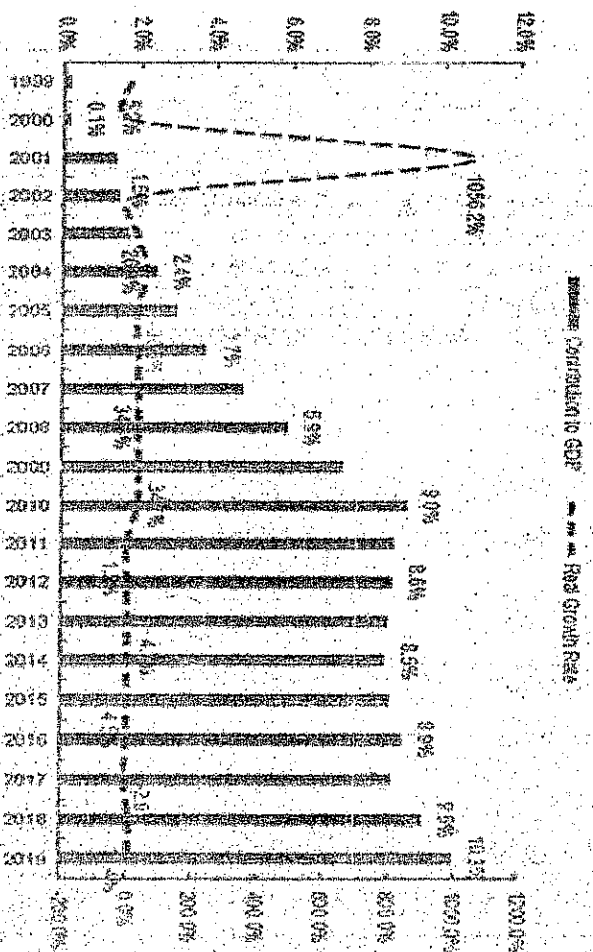


Figure 1.6 The Nigerian Telecommunications Sector Report: A Transformative Past, Resilient Future

1.4.8 Additional Uses

New applications are becoming feasible as network connectivity and wireless devices become more widely available, as well as breakthroughs in affiliated electronics technologies. Mobile SMS communication has become extremely popular. The combination of cameras and Wi-Fi allows for some type of image communication. Cellular carriers are installing location determination features (at the request of the FCC, for security purposes), with some employing the global positioning system (GPS).

When these features are all in place, a plethora of new applications other than public safety will be obtainable. Numerous people believe that electronic gaming is a tremendously thriving industry that is still in its early life.

1.5 Conclusion

The long record of advancement in wireless and RF communications has resulted in a plethora of technologies suitable for a wide variety of applications. We are living in an era of rapid advancement in wireless communication. When combined with the currently added Online

material, the door has become open for new applications ranging from telemetry to multimedia. Mobile phone services proliferated in the 2000s; the mobile broadband revolution will become even more active now that the sorts of equipment and information are considerably more diverse.

Exercises

- 1) What exactly does the wireless transmission idea entail?
- 2) Explain how the wireless landscape is inhabited with respect to fixed location.
- 3) We may think of wireless goods in terms of the capabilities they support, such as telephony, messaging, private data, and Internet service. Explain
- 4) Why has the Internet's explosive growth, especially its digital goods and services, created new obstacles for the development of mobile communications networks?
- 5) Explain the concept Software or Cognitive Radio